Aerodynamics Parameters Prediction Using Convolutional Neural Network (CNN)

Kavyashree Nagaraju Soumya Vardhana Gari
Krishna Parampalli Mogaveera Shivani Podila
Priyanka Kaparaboina

Department of information science, University of North Texas, Texas, USA
February 25, 2024

1 Abstract

The method of Convolutional Neural Networks (CNN) is primarily used for Visual analysis purpose, constitute a category of deep neural networks for meta-modeling tasks of aerodynamic such as image classification, object detection, and image segmentation. Based on the dataset with adequate quantity, the aerodynamic property of airfoil can be predicted using CNN. To find the relationship between lift-to-drag ratio and air-foil shape through neural network, the first and critical step is preparing the input data with the proper mathematical representation for training the neural network. The datasets of NACA series airfoils are downloaded from UIUC airfoil. Thousands of coordinates will be imported and generated in different software for analysis. The three main goals of this project are to predict Aerodynamic parameter of airfoil such as lift-to-drag ratio (Cl/Cd) with various values of alpha angles of attack; calculate the efficiency and the accuracy of airfoil; Comparing the respective CNN method versus other software such as CFD (Computational fluid dynamics)or XFLR5 as well as Theoretically (Numerical method). It even shows that CNN is 5000 times more accurate and highly efficient than numerical method or other software method (CFD). The targeted result of this project focuses three main aspects are training and validation loss with respect to epoch; Test and predicted confusion matrix;

comparison of predicted value & truth value, time consumption table of other software, accuracy, as well as efficiency of model. Furthermore, the CNN's ability to generalize across different geometries and flow conditions is evaluated, showcasing its versatility and applicability across various aerodynamic scenarios.

Keywords: Convolutional Neural Networks (CNN), Visual Classification, Datasets, Airfoil, Computational Fluid Dynamic (CFD), lift-to-drag ratio (Cl/Cd), alpha (Angles of attack), Numerical Method.

2 Introduction

Aviation field has tons of activities to work in Designing, Analyzing, manufacturing, Maintenance, and performance. When there are tasks involved in inventing the design of the aircraft On the other hand, there are thousands of airplanes in MRO (Maintenance, Repair, and Operations) for best performance. Whereas a tiny negligence of 0.05mm dent can also cause a risk and therefore remained in a MRO. Traditionally, analysis of aerodynamics is relied on CFD (Computational Fluid Dynamics) Simulation (using CFD, XLFR5, Ansys software) and, Wind tunnel testing moreover theoretically it is managed by solving PDE (partial differential equations). Airfoil is a streamlined shape within the wings, it is just as equal to cross section of the wings of airplane, they are followed up by many NACA 4-digit series containing nearly 1600 of coordinates.

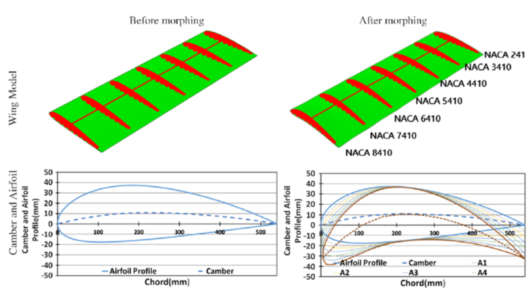

Figure 1: Airfoil profile within wings showing 4-digit NACA series

CNN is nothing but Convolutional Neural Network used in Aerodynamic meta-modelling tasks usually to predict the parameters of airfoil such as lift-to-drag ratio (cl/cd) with different angles of attach (alpha) which determines

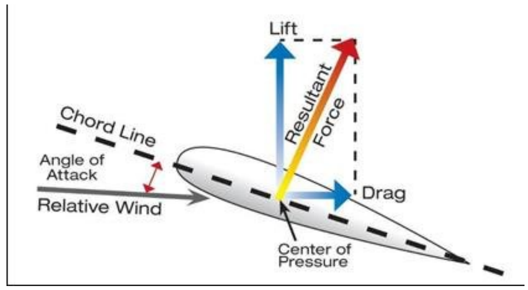

Figure 2: Aerodynamic Forces

the accuracy as well as has better efficiency comparing to analytical method and CFD method as they are time consuming too. CNN is usually used by extracting features of figures and images, and then imported in software to generate figures of contours containing coordinates data in the form of airfoil with particular Pixels size as well as set of panels to obtain maximum data or properties such as laminar flow, turbulent flow, cl/cd and various predictions of it.

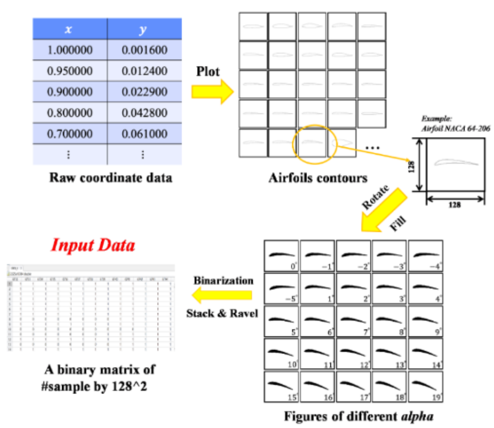

Figure 3: Image generating Input dataset creating

Figure 3: Image generating input dataset

Machine learning method is widely used to solve problem in computer science and aerodynamics field using various machine learning algorithms. Just alike CNN, [6], Olayemia et al. MLP (Multi-layer perceptions) is also used to predict aerodynamic parameter and providing a faster and more efficient as usual. MLP is specially used to achieve two-dimensional aerodynamic design under framework of classification and regression. [6] presents a data-driven model using CNNs and MLPs to predict aerodynamics lift coefficients of supercritical airfoils in transonic flow regimes. The study demonstrates the potential for faster and more efficient prediction of lift coefficients compared to traditional methods. Whereas when it comes to Deep Convolutional Neural Network [9], Yan Wu et al. research paper presents a CNN-DCNN model for predicting flow fields around airfoils, utilizing SDF representations and training parameters optimization. The model demonstrates high prediction accuracy (up to 98.23 percentage for pressure and velocity fields) and significant speed improvement (three orders of magnitude faster than CFD simulations). It [12], Bo-Wen Zan et al. introduces a CNN-based method for high-dimensional aerodynamic modeling to address the "curse of dimensionality" and applies it to surrogate-based aerodynamic shape optimization. By creating a data-driven method [3], Hai Chen et al. that employs CNNs to predict multiple aerodynamic coefficients, including lift, drag, and moment coefficients across a range of airfoil designs, the work improves the field of aerodynamics applications. Offers a improved accuracy and efficiency in the analysis and design optimization of airfoils. In order to solve the problem of accurately predicting airfoil lift coefficients, the authors [11], Boping Yu et al. modified the CNN design. They demonstrate the implementation of the mini-batch SGD, dropout, and BN layers. Important aspects included are preparation techniques for airfoil data and optimized hyper-parameters. [14], Yanxuan Zhao et al. providing an example of a visually understandable CNN architecture for predicting the aerodynamic coefficients, this work advances the field of aeronautical engineering. Their primary area of study was 2D airfoils. CNN-based flow control device modelling on aerodynamic airfoils method [7], Koldo et al. demonstrates that neural networks may be used to investigate flow control devices with tolerable faults and a significant decrease in the amount of computational time and resources needed. Using deep learning approaches, flow control features were implemented in airfoils as part of the effort. There was another new method [2], J Fidkowski et al. to perform output error estimation and mesh adaptation in computational fluid dynamics (CFD) using CNN. This technique captures both the local and global features associated with the numerical error using an encoder-decoder type convolutional neural network (CNN). [4], Min Il Kim et al. In order to generate the precise surface distributions of Cp as the dataset, it numerically simulated a steady and viscous 2D flow over the airfoil. Lift to drag ratio and pressure fields were obtained by numerically simulating all flows around the airfoils predicted by the established CNN model. [8], Yuqi Wang et al. explains the application of Dual Convolutional Neural Networks (Dual-CNN) to wind turbine optimization in aerospace engines is covered in this research. It demonstrates that in terms of efficacy and torque forecasting, Dual-CNN performs better than conventional techniques such as Gaussian process regression (GPR) and Artificial Neural Network (ANN). In [15], Yong Zhao et al. a circular cylinder's pressure and airflow are examined using two CNN models. While constrained flow is handled by CFD modeling, outer boundary information is obtained using Euclidean measurements.Through multi-point optimization, [1], Buckley et al. the study seeks to balance design restrictions for practical aerodynamic problems. It emphasizes how crucial it is to precisely evaluate design issues in order to guarantee peak performance. Article [5], Mathias et al. explores the estimation of aerodynamic characteristics for non-conventional airfoils using Convolutional Neural Networks (CNNs). It offers information about the particular CNN architecture that was utilized, the training process, and the assessment measures that were applied to determine how accurate the model's predictions were. Article [13],Zhang et al. The investigators utilize a multi-objective optimization methodology to optimize aerodynamic performance while taking into account several variables at once. By offering insights into increasing aerodynamic efficiency—which is essential for boosting train performance and lowering energy consumption. [10], Yao et al. A technique for optimizing the three-dimensional aerodynamic design of high-speed train noses.This study intends to improve high-speed train aerodynamic performance by combining these methodologies, leading to advances in train design for increased speed, efficiency, and safety.

3 Dataset

Dataset of airfoil are usually downloaded from the UIUC Airfoil Dataset which is a raw data containing coordinates with nearly 1600 mainstream airfoils, ranging from the NACA 4-digit series to the Selig series. The entire raw data is taken into an account in the software to generate the proper contour figures whereas aerodynamics parameters is predicted such as lift coefficient of airfoil, Angle of attack , Reynolds number (Re) as well as Mach number (M a). The dataset, that is composed up of 20,578 photos from 1,000 airfoil samples, is utilized to train and assess the CNN model. In one of the studies, a collection of pictures created from airfoil samples with different angles of attack and whereas resulting in each airfoil sample that is having a total of 100 pictures. For the best efficiency of aerodynamics behavior, the lift-to-drag ratio is an essential aerodynamics parameter to predict instead of just the lift coefficient. The dimension of the image is set to (128,128) rather than the more common smaller sizes, like (28, 28) in the MNIST dataset or (32, 32) in the NIST36 dataset and this increment in resolution improves the model's sensitivity and accuracy in predicting aerodynamic attributes by containing more information and enable the $1\circ$ (Angle of attack) increment. This is because the aerodynamic performance of airfoils is extremely sensitive to even small changes in their shape, and maintaining a reasonable resolution is crucial to maintaining prediction accuracy.

4 Methods

The methodology focuses to evaluate the CNN model's performance by predicting an airfoil's lift-to-drag ratio from images generated by different angles of attack of sample airfoil. Thereby the dataset is separated into training and testing sets in the second step to ensure the CNN) model's robustness and generalization, whereas to maximize training speed and accuracy. The methodology also uses batch normalization in the CNN architecture. The testing set is utilized to evaluate the model's ability to be applied to new data, whereas the training set is often used to train the model. The CNN algorithm is trained on the underlying patterns in the data using the training dataset, which made up 70 percentage of all the samples. To avoid overfitting and improve the model, 20 percentage of the samples were chosen from the validation dataset and remaining 10 percentage of the samples formed the testing dataset, which was kept apart and is used to assess how well the model performed on unobserved data. The neural network's overall efficiency is increased by implementing batch normalization in each convolutional layer, which leads to faster convergence and better generalization performance for the model. Applied strategy stands out because it makes use of improved image resolution, predicts the lift-to-drag ratio, and integrates batch normalization methods into the CNN architecture. Various preprocessing techniques, data manipulation methods, and augmentation strategies were employed to adequately prepare the dataset for training the model. The objective behind these approaches was to elevate the data's quality, optimize model performance, and mitigate overfitting.

4.1 Theoretical Method

The main part of the analytical method in the airfoil calculation is the Navier-Stokes Equation which is a partial differential equation. The Navier-Stokes equations describe the motion of fluid substances and are one of the pillars of fluid dynamics. For an incompressible, viscous fluid flow, the Navier-Stokes equations can be simplified. The incompressibility condition means that the density () of the fluid is constant, and this leads to the continuity equation: u=0 where u is the velocity field of the fluid For an incompressible Newtonian fluid, the Navier-Stokes equations under the assumption of constant viscosity () and without external forces can be written as: (t/u+uu)=p+^2u

In a more detailed form, considering a three-dimensional Cartesian coordinate system, the incompressible Navier-Stokes equations can be expressed for each velocity component (u, v, and w) as follows: (u/t +u u/x+vu/y +wu/z) = p/x +($^2u/x^2 +^2 u/y^2 +^2 u/z^2$)($v/t + uv/x + vv/y + wv/z$) = $p/y+(^2v/x2+^2v/y^2+^2v/z^2)(w/t+uw/x+vw/y+ww/z) = p/z+(^2w/x^2+^2 w/y^2+^2 w/z^2) consideringthecontinuity : (u/t$

Getting an approximate solution of this equation is much more feasible, but the process is still time-consuming and extremely inconvenient.

4.2 Computational Fluid Dynamics (CFD Method)

CFD represents a numerical approach employed to model the dynamics of fluid flow around various entities, including airfoils. By solving the fundamental equations governing fluid motion, CFD facilitates the anticipation of aerodynamic characteristics such as lift and drag forces, pressure distribution, and flow patterns, obviating the necessity for traditional wind tunnel experiments.

Calculating parameters of airfoil can be solved by vortex panel method which basically focus on getting numerical result that can approximate the analytical solution. The main steps involved in calculating the parameters of an airfoil using the vortex panel method are Defining the Airfoil Geometry, Computing the Influence Coefficients, Enforcing the No-Penetration Boundary Condition, Solving the Panel Vortex Strength, Calculating the Surface Pressure Distribution and then Calculate the Lift Coefficient.Through CFD

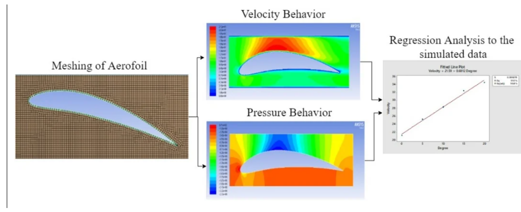

Figure 4: Image of Computational fluid dynamics analysics

simulations, detailed aerodynamic data can be acquired, allowing researchers to verify the CNN model's precision in predicting aerodynamic characteristics.

4.3 Convolutional Neural Network (CNN)

This technique involves predicting the lift-to-drag ratio of airfoils which could significantly speed up the design cycle for new airfoils by providing a quick estimate of aerodynamic performance without the need for expensive and time-consuming simulations or wind tunnel testing for each design iteration. This prediction was based on images derived from airfoil samples with different angles of attack.

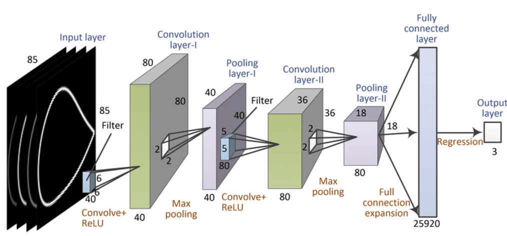

Figure 5: Process of Convolutional Neural Network

CNN is recognized for its capability to automatically acquire hierarchical features from input data and is frequently employed in image processing tasks. The process involved in CNN are Architecture, Training, Loss function, Activation functions, Regularization, and Evaluation. The main goal of using CNN is to create a predictive model. This model can be used to accurately predict the lift-to-drag ratio of airfoils by analyzing their visual representations which will demonstrate the effectiveness of employing deep learning techniques in aerodynamics research.

5 Experiment

Usually, CNN software is to predict lift-to-drag ratio. To prove this in certain accurate and efficient, mean squared error, confusion matrix and time consumption table is been used. Thousands of airfoil samples are considered and split into 70 percentage of training datasets and 20 percentage of testing datasets. The range of angle of attack is set with certain degree of increment. All these Training datasets and testing datasets stages performed by using Pytorch and PyCharm, it is one of the most popular choice, and is an IDE (Integrated Development Environment).

5.1 Code Explanation

Initially, File Existence and Integrity Check is used, using the OS module, it retrieves the permissions of the necessary data file and determines if it is empty or not. Also Preprocessing and Data Loading uses scipy.io.loadmat() to load the data from a MATLAB file. Hereby, the data is divided into sets for testing, validation, and training. Definition of a Convolutional Neural Network: torch.nn.Module is used to define a CNN model architecture. The model is composed of convolutional layers, fully connected layers, batch normalization, max-pooling, and ReLU activation functions. Training Environment Setup: It defines the optimizer (Adam), the loss function (Mean Squared Error), the device (CPU or GPU), and moves the model to the device. Data Loader Creation: Torch.utils.data is used to generate data loaders. Batches of training, validation, and test data are handled by DataLoader. Training Loop: A predetermined number of epochs are used in the training loop. The model is trained with the training data within each epoch, and the loss is computed. The model's performance is assessed on the validation set at the end of each epoch. Training and Validation Loss Visualization: Using matplotlib.pyplot, the script depicts the training and validation loss over epochs. Understanding how the model performs during training and if it overfits or underfits is made easier with the aid of this graphic. Testing and Results Visualization: The testing set is used to test the trained model, and the predicted and actual Cl/Cd ratios are compared. Additionally, the testing duration is shown. Lastly, a plot of the actual and predicted values is used to compare the model's performance visually.

This shows how to use PyTorch for data management and visualization in the process of creating, training, and assessing a CNN model for a particular prediction problem.

6 Result and Discussion

Following the construction of the CNN model and data pre-processing, training is carried out, and generating a number of training outcomes. The confusion matrix, training and validation loss, accuracy, and efficiency of the model are the three primary areas of focus for the outcomes.

Code results are in the following:

The File Existence and Integrity Check shows that the file exists by not printing either of the statement loop such as “File does not exist” and “File is empty”. Further, mentions the File permission as “0o100644”. After data is load, it is split into training, testing datasets and validation with 70, 20 and 10 percentages respectively. 30 Epochs are resulted and the resulted graph represents the Loss and Epoch in X and Y axis respectively representing Training and validation loss over Epochs. Testing time also reports as that testing completed in 4.225805 seconds. The purpose of this validation datasets is that during time these training losses and validation losses is compared. Based on the convergence of the training and validation loss: The training loss converged starting from epoch 12 and the validation loss converged starting from epoch 8. Whereas the training loss is approximately 0.6 and the validation loss is approximately 0.1 at the final epoch, which is 30.

Eventually, for Predicted vs Actual Cl/Cd Ratios displays a comparison of the ground truth and the prediction. The ground truth, CL/CD of each airfoil sample at a single angle determined by CFD is shown by orange lines. The predictions made by our CNN model are shown as blue lines. Lift-to-drag ratio is the y-axis, while foil serial number is the x-axis. Overlapping of these two plots represents that our prediction accurately reflects the ground truth. There is a deviation at the epoch value of 100. Typically, Epochs are used in machine learning to quantify the number of times a complete pass through the training data is made; on the other hand, deviation may indicate the moment at which the validation loss and training loss start to diverge.

7 Conclusion

The three main goals of this project were to predict the Aerodynamic parameters of airfoils such as lift-to-drag ratio (Cl/Cd) with various values of alpha angles of attack; calculate the efficiency and the accuracy of airfoils; Compare the respective CNN method versus other software such as CFD. 1. As the resultant graph represents the overlapping of actual values and predicted values further it defines that our prediction value accurately reflects the ground truth. 2. Efficiency: The CFD method's solver iterates over each sample and performs several computations until the results converge. Accuracy: Evaluating the performance of the model using metrics like Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE), which measure the difference between the predicted and actual values results in accuracy. 3. On the other hand, our CNN model will only go through all samples once following fine-tuning and training, resulting in a linear increase in processing time. The CNN method outperforms the CFD method by 5000 times when the testing sample size is 300.

References

[1] Howard P Buckley, Beckett Y Zhou, and David W Zingg. Airfoil optimization using practical aerodynamic design requirements. *Journal of Aircraft*, 47(5):1707–1719, 2010.

[2] Guodong Chen and Krzysztof J Fidkowski. Output-based adaptive aerodynamic simulations using convolutional neural networks. *Computers & Fluids*, 223:104947, 2021.

[3] Hai Chen, Lei He, Weiqi Qian, and Song Wang. Multiple aerodynamic coefficient prediction of airfoils using a convolutional neural network. *Symmetry*, 12(4):544, 2020.

[4] Min Il Kim and Hyun Sik Yoon. Geometric modification for the enhancement of an airfoil performance using deep cnn. *Ocean Engineering*, 266:113000, 2022.

[5] Marlon Mathias and Marcello Medeiros. Interaction between rossiter and görtler modes in the compressible flow in an open cavity. In *Aiaa Aviation 2020 Forum*, page 3074, 2020.

[6] Olalekan Adebayo Olayemia, Oluwadolapo Isaac Salakoa, Abdulbaqi Jinadu, and Adebowale Martins. Aerodynamic lift coefficient prediction of supercritical airfoils at transonic flow regime using convolutional neural networks (cnns) and multi-layer perceptions (mlps). *Al-Qadisiyah Journal for Engineering Sciences*, 16(2), 2023.

[7] Koldo Portal-Porras, Unai Fernandez-Gamiz, Ekaitz Zulueta, Alejandro Ballesteros-Coll, and Asier Zulueta. Cnn-based flow control device modelling on aerodynamic airfoils. *Scientific Reports*, 12(1):8205, 2022.

[8] Yuqi Wang, Tianyuan Liu, Di Zhang, and Yonghui Xie. Dual-convolutional neural network based aerodynamic prediction and multi-objective optimization of a compact turbine rotor. *Aerospace Science and Technology*, 116:106869, 2021.

[9] Ming-Yu Wu, Yan Wu, Xin-Yi Yuan, Zhi-Hua Chen, Wei-Tao Wu, and Nadine Aubry. Fast prediction of flow field around airfoils based on deep convolutional neural network. *Applied Sciences*, 12(23):12075, 2022.

[10] ShuanBao Yao, DiLong Guo, and GuoWei Yang. Three-dimensional aerodynamic optimization design of high-speed train nose based on ga-grnn. *Science China Technological Sciences*, 55:3118–3130, 2012.

[11] Boping Yu, Liang Xie, and Fuxin Wang. An improved deep convolutional neural network to predict airfoil lift coefficient. In *International Conference on Aerospace System Science and Engineering*, pages 275–286. Springer, 2019.

[12] Bo-Wen Zan, Zhong-Hua Han, Chen-Zhou Xu, Ming-Qi Liu, and Wen-Zheng Wang. High-dimensional aerodynamic data modeling using a machine learning method based on a convolutional neural network. *Advances in Aerodynamics*, 4(1):39, 2022.

[13] Liang Zhang, Ji-ye Zhang, Tian Li, and Ya-dong Zhang. . *Journal of Zhejiang University-SCIENCE A*, 18:841–854, 2017.

[14] Yanxuan Zhao, Chengwen Zhong, Fang Wang, Yueqing Wang, et al. Visual explainable convolutional neural network for aerodynamic coefficient prediction. *International Journal of Aerospace Engineering*, 2022, 2022.

[15] Yong Zhao, Xuehui Jiang Zhao, and Yang Meng. Prediction of confined flow field around a circular cylinder and its force based on convolution neural network. *IEEE Access*, 10:6889–6898, 2021.